Assessing approaches to measuring interpersonal physiological synchrony between dancers and individuals with dementia

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Background

- Over the next 30 years, the number of people living with dementia in Canada is expected to grow by 187% [1]
- Importance of approaches in dementia care that support personhood and honor relations
- Measure the lingering effects of *Mouvement de Passage* on the embodied memory of older adults with dementia
- Interpersonal physiological synchrony (IPS) characterizes the dynamic interaction of physiological responses between individuals[2]
- The dominant IPS approaches are concurrent[3] and time-lagged[4]
- Physiological responses of older adults with dementia living in CHSLDs are attenuated with a limited dynamic range

Fig. 1. *Mouvement de Passage* is a program that fosters connection with residents of long-term care homes through movement.

Objective

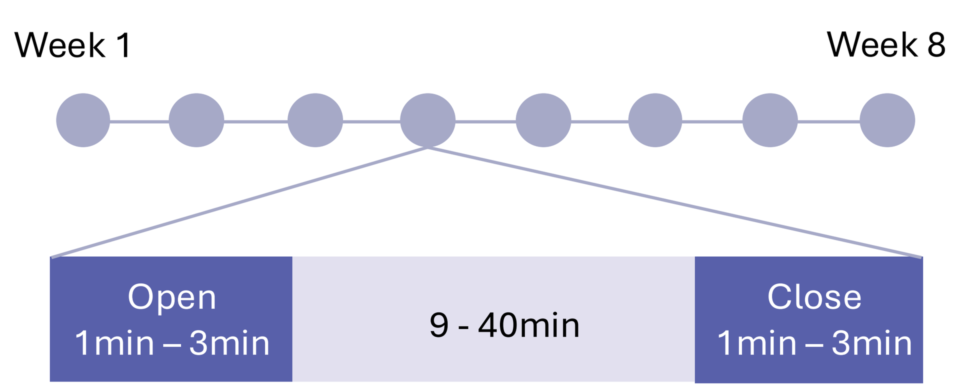
Evaluate the feasibility of applying an established IPS pipeline, developed for healthy populations, to resident-dancer dyads in *Mouvement de Passage*

→ Adapt synchrony pipeline to account for the altered physiological patterns of residents

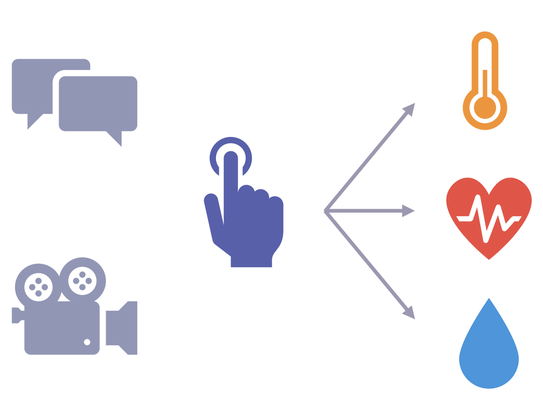
Materials and Methods

Study Design

- N = 4 resident-dancer dyads (8 participants)
- 8 week longitudinal case series study
- Analyze opening and closing moments of each session



Data Collected

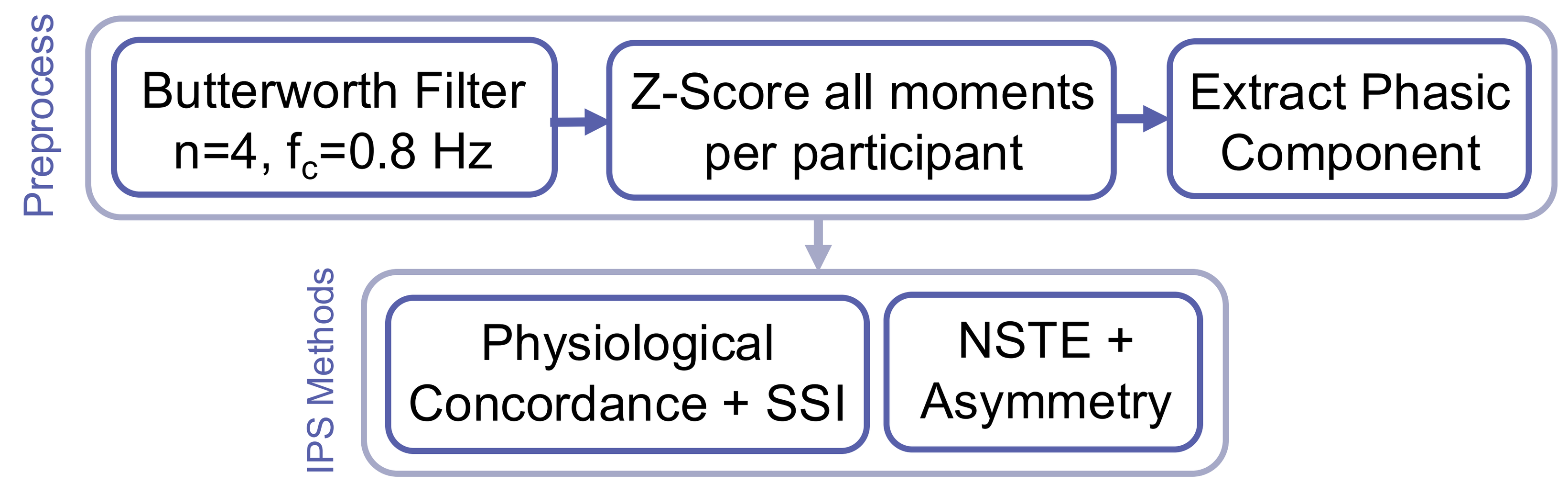


Skin temperature

Blood volume pulse (BVP)

Electrodermal activity (EDA)

Pipeline



Preprocess

Filtering

Approach: Adapted from 1 Euro filter to Butterworth to better preserve signal features

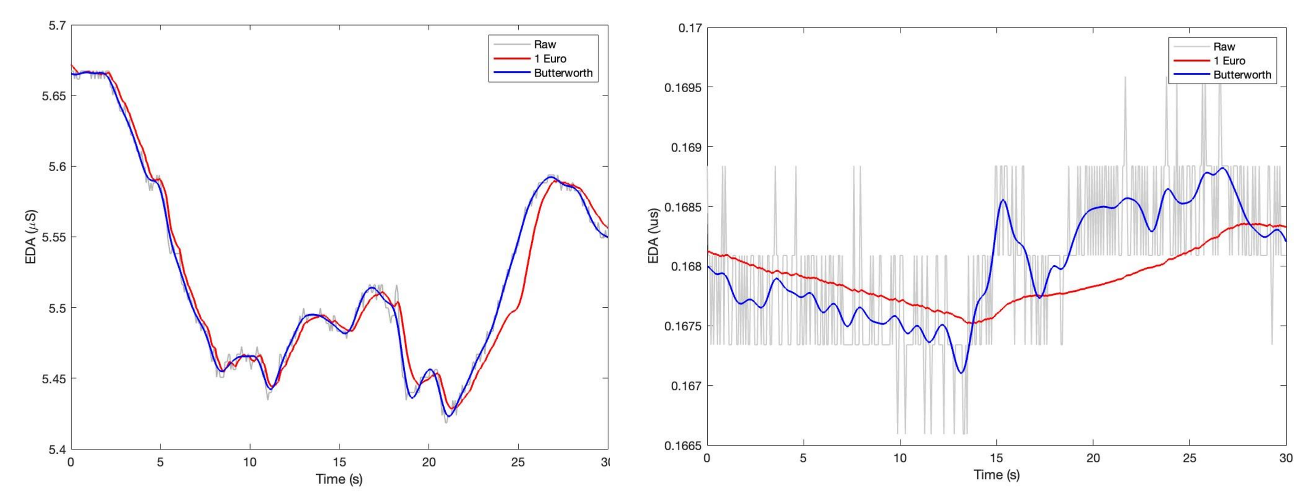


Fig. 2. Comparison of 1€ ($f_{C,min}$=1, β=10) and Butterworth (n=4, f_C=0.8) filter on dancer (left) and resident (right) signals.

Normalization

Approach: Z-score across all moments per participant to enable meaningful signal comparison across sessions and between dancers and residents

Decomposition

Approach: Extract phasic component using cvxEDA toolbox to remove the slow drift in EDA and satisfy assumption of stationarity for IPS methods

EDA IPS Methods

Physiological Concordance: Moment-to-moment synchrony

1. Average Slope: 5s window, 1s step
2. Rolling Pearson Correlation: 15s window, 1s step, +/- 8s lag
3. Area Under the Curve per lag → Max AUC = Max Time Lag
4. Single Session Index (SSI): over entire segment

$$SSI = ln\frac{\sum_{R>0} R}{\sum_{R<0} |R|}$$

Normalized Symbolic Transfer Entropy: Direction of information transfer

Parameters: embedding dimension, time delay, prediction time

$$TE_{X\rightarrow Y} = I(Y^F;X^P|Y^P) = H(Y^F;Y^P) - H(Y^F;X^P,Y^P)$$

Preliminary Results

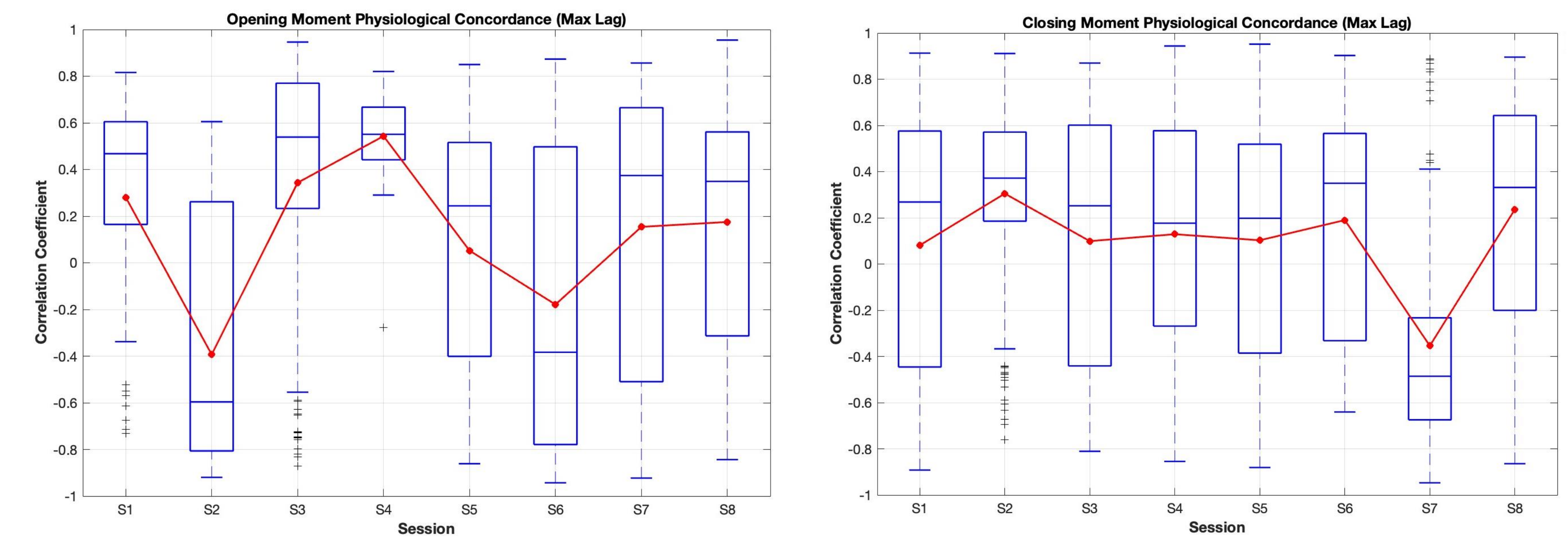


Fig. 3. Physiological Concordance of opening (left) and closing (right) moments over 8 sessions for Dyad 1.

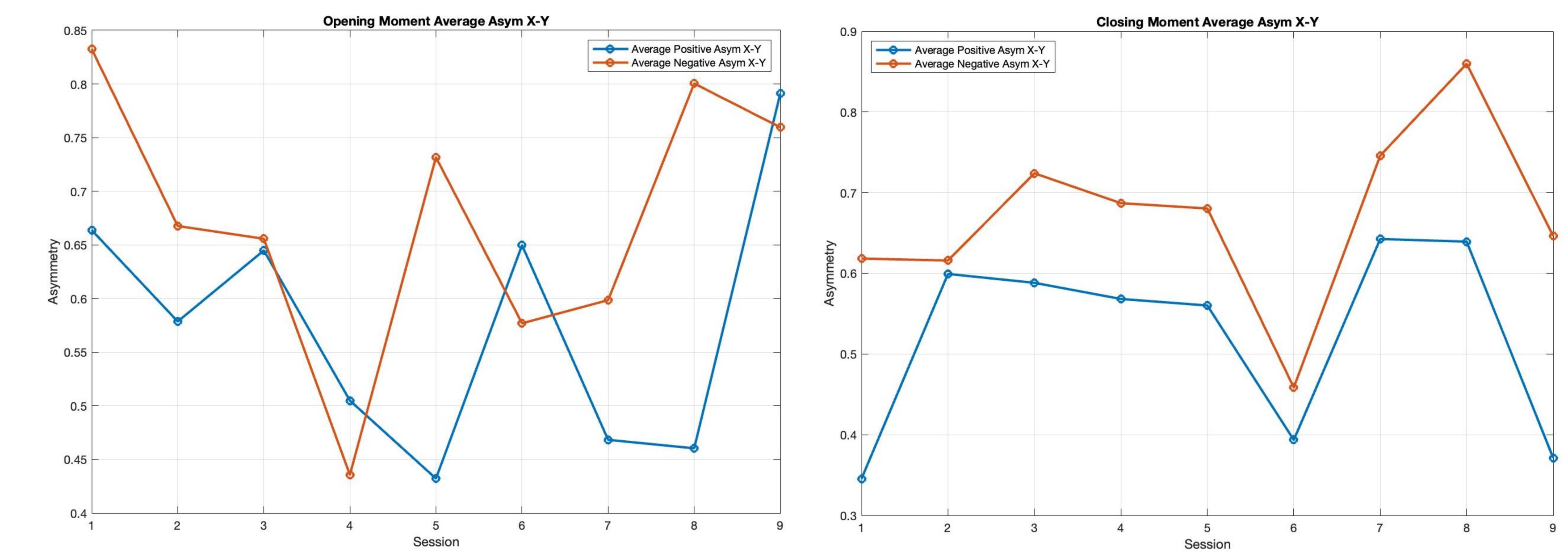


Fig. 4. Average magnitude of positive and negative asymmetry of X (resident) to Y (dancer) of opening (left) and closing (right) moments over 8 sessions for Dyad 1.

Conclusions/Future Works

- Adapting the preprocessing and synchrony methods to the resident data and study design enabled more accurate and meaningful measures of IPS.
- Once the pipeline is applied to all dyads, additional techniques will be explored for result interpretation.
- Future work involves adapting the pipeline for skin temperature by considering its slower response time. Additionally, other IPS methods should be explored for BVP analysis.

Acknowledgements

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1 Armstrong, J., Sivananthan, S. & Guimond, J. THE MANY FACES OF DEMENTIA IN CANADA: EXAMINING SEX, ETHNICITY, AND AGE OF ONSET IN THE LANDMARK STUDY. *Innovation in Aging* **7**, 88 (2023). 2 Fu, D., Incio-Serra, N., Motta-Ochoa, R. & Blain-Moraes, S. Interpersonal physiological synchrony for detecting moments of connection in persons with dementia: A pilot study. *Frontiers in psychology* **12**, 749710 (2021). 3 Marci, C. D., Ham, J., Moran, E. & Orr, S. P. Physiologic correlates of perceived therapist empathy and social-emotional process during psychotherapy. *The Journal of nervous and mental disease* **195**, 103-111 (2007). 4 Lee, U., Blain-Moraes, S. & Mashour, G. A. Assessing levels of consciousness with symbolic analysis. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* **373**, 20140117 (2015).